Dynamic Planet C - Hydrology

Lexington Invitational 2022



Jaime Yu, David Zhou

INSTRUCTIONS

Hello! Welcome to the LexSO Dynamic Planet test written by David Zhou and Jaime Yu (Hoatzin 4913) from Lexington High School.

- You have 50 minutes to complete this test
- The Test is split up into 9 stations. I recommend that you do each station in order, but you can take the stations in any order you want
- **This test contains 98 questions and 215 points**
- Show work for calculation questions where a text box is provided
- There is a **supplemental image sheet included in the folder. YOU MUST USE THIS TO DO WELL**
- A **feedback form** is provided at the end of the test. Please fill it out if you get a chance!
- Please, don't cheat - it's not cool >:C
- As a note, some point values and questions have been changed since the test was administered for higher accuracy. It is roughly 99% the same.

1 Water Budget and Water Use

The following 4 questions will be based on the Charles River watershed which has an area of approximately 800 km^2. For now, you may assume the following facts:

- Only input of water into watershed is through precipitation
- 40% of all precipitated water infiltrates as groundwater
- This region receives an average of 45 cm of rain annually
- All other outflows of water in this watershed is from evapotranspiration and river runoff from the Charles River
- The Charles river has an approximate discharge of 9 cumecs. 3 cumecs are fed by surface runoff.
  - 3 cumecs are fed by surface runoff
  - 2 cumecs are fed by a stream which gets water from outside the Charles Watershed
  - the remainder is fed by groundwater

1. (5 points) Calculate the total amount of water that returns to the atmosphere via evapotranspiration in a year in cubic meters. Provide your answer in scientific notation with 3 sig figs and show your work.

   Solution: $0.6 \cdot 0.45 \cdot 800 \cdot 1000^2 = 2.16 \cdot 10^8 m^3$ input that does not flow into the ground
   $3 \cdot 365 \cdot 24 \cdot 60^2 = 9.46 \cdot 10^7 m^3$ outflow from charles from runoff
   $2.16E8 - 9.46E7 = 1.214 \cdot 10^8 m^3$ water that leaves via Evapotranspiration

2. (2 points) Calculate the amount of water that infiltrates into the ground as groundwater annually in cubic meters. Provide your answer in scientific notation with 3 sig figs and show your work.

   Solution: $0.4 \cdot 0.45 \cdot 800 \cdot 1000^2 = 1.44 \cdot 10^8 m^3$

3. (2 points) The Charles river is which of the following? Select all the apply.

   A. A gaining stream

   B. A loosing stream

   C. A disconnected stream

   D. A connected stream

4. (3 points) Does the Charles River watershed recharge or discharge its groundwater supply? By how many cubic meters each year? Provide your numerical answer in scientific notation with 3 sig figs. Show your work.

   Solution: $9 - (2 + 3) = 4 cumecs$ outflow of charles that is fed by groundwater
   $4 \cdot 365 \cdot 24 \cdot 60^2 = 1.26 \cdot 10^8 m^3$ outflow from charles from groundwater
   $1.44 \cdot 10^8 - 1.26 \cdot 10^8 = 1.79 \cdot 10^7$ net groundwater change annually in cubic meters
   Because the net change is positive, it recharges its groundwater supply

For the following question:
P = Amount of precipitation
R = amount of run-off
E = amount of evaporation
T = amount of transpiration
I = amount of infiltration

5. (1 point) Which of the following is typically true?
   A. $E > T$
   B. $P = E + T$
   C. $T + E > I$
   D. $T + R + E + I > P$

6. (1 point) Which of the following holds the most freshwater?
   A. Lakes
   B. Glaciers
   C. Groundwater
   D. Permafrost

7. (1 point) Which of the following is the correct order from least salty to saltiest water?
   A. Freshwater, Brackish water, Briny Water, Salt water
   B. Freshwater, briny water, brackish water, saltwater
   C. Freshwater, brackish water, salt water, briny water
   D. Freshwater, briny water, saltwater, brackish water

8. (1 point) Which of the following is closest to the amount of water found in glaciers?
   A. $2.4 \cdot 10^9 km^3$
   B. $2.4 \cdot 10^8 km^3$
   C. $2.4 \cdot 10^7 km^3$
   D. $2.4 \cdot 10^6 km^3$

9. (1 point) Which of the following composes the highest percentage of human groundwater use in the world?
   A. Domestic
   B. Agriculture
   C. Industrial
   D. Mining

10. (1 point) What is the name of the process in which water condenses into water vapor straight from ice?
    A. Sublimation
    B. Deposition
    C. Condensation
    D. Vaporization

11. (3 points) List 3 negative ecological effects of dams.

Solution: Anything reasonable. Can include: thermal pollution, downstream downcutting, armouring, prevention of fish migration, no more floodwater recharge for nutrients

12. (2 points) Answer the following questions on artificial levees:
    (a) (1 point) What are artificial levees and why are they built?
    (b) (1 point) How do they affect flooding in the river where they are build along? Why is this?

Solution: a) Human-made sediment walls along a river. Built to prevent or maybe even stop flooding.
b) Water levels are higher and water flows faster in floods because water is forced to go through a tighter space because of levees

13. (1 point) About what percent of lakes have nutrient pollution?
    A. 10%
    B. 35%
    C. 50%
    D. 75%

14. (2 points) What are algal blooms, and how to human impacts cause them?

Solution: Algae blooms are when there is a lot of growth of algae on a water surface. Human impacts can create these blooms when fertilizer run-off along rivers or streams add excess nutrients to water bodies so algae grows explosively

15. (2 ½ points) Random lake trivia time, cuz that belongs on every DP test this year :D
    (a) (½ point) What is the deepest lake in the world?
    (b) (½ point) What is the longest river in the world?
    (c) (½ point) What is the most polluted lake in the world?
    (d) (½ point) What is the largest endoheic lake in the world?
    (e) (½ point) What is the largest waterfall by volume?

Solution: a) Lake Baikal b) Nile River c) Lake Karachay d) Caspian Sea e) Inga Falls

2 Stream Geology Round 1

1. (2 points) Refer to Station 2 Image A which is a topographic image of part of Campbell Creek. In which direction is Campbell Creek flowing towards and how do you know?

   Solution: East (1pt), the V's open up in the direction of flow (1pt)

2. (1 point) Campbell Creek is a
   A. perennial stream
   B. intermittent stream
   C. disappearing stream
   D. braided river

3. (2 points) *Select all of the following which are likely true regarding Campbell Creek
   A. it is connected to the water table
   B. it has a gravelly bed
   C. it has a turbulent flow
   D. it has a larger hydraulic radius
   E. it has a steep gradient
   F. it can be considered a headwater tributary

4. (1 point) Refer to Station 2 Image B which is a topographic map of the Juniata River. What type of river channel does this reach of the Juniata River display?
   A. straight
   B. anastomosing
   C. meandering
   D. braided

5. (1 point) True or False: This section of the Juniata river is considered "mature"

6. (1 point) True or False: The Juniata River has a higher capacity, competence, velocity and discharge than Campbell Creek.

7. (2 points) What term is used for the region shaded in light purple surrounding the Juniata River?

   Solution: floodplain

8. (5 points) Refer to Station 2 Image B which is a topographic map of the Susquehanna River where the Juniata River connects as a tributary. Please answer the following questions regarding this river.
   (a) (1 point) Is it a superposed river or an antecedent river?
   (b) (3 points) Justify your response to the question above
   (c) (1 point) Did Peters Mountain (black arrow) form first, or Susquehanna River?

Solution: a) Antecedent
b)The stream cuts through the ridges (1pt), which means that it must have first been established or else if would have gone around the ridges (1pt)
c)Susquehanna River

9. (1 point) Which of the following is true concerning the bars in the Susquehanna River as seen from the satellite image on the right.
   A. The braided index (total bar length/reach length) is less than 2 so thus the Susquehanna River is not considered braided
   B. The braided index (total bar length/reach length) is over 2 so thus the Susquehanna River is considered braided
   C. They do not migrate and thus Susquehanna River is not considered braided
   D. They do not migrate due to large amounts of vegetation and thus Susquehanna River is not considered braided

10. (1 point) Which of the following correctly orders the rivers from lowest to highest stream order using the Strahler ordering system?
   A. Campbell Creek, Juniata River, Susquehanna River
   B. Campbell Creek, Susquehanna River, Juniata River
   C. Juniata River, Susquehanna River, Campbell Creek
   D. Susquehanna River, Juniata River, Campbell Creek

11. (1 point) Refer to Station 2 Image D. This is a topographic map of region D which contains a river and some tributaries. This is an example of which drainage pattern?
   A. annular
   B. dendritic
   C. parallel
   D. rectangular
   E. radial
   F. trellis

12. (1 point) Refer to Station 2 Image E. This is a topographic map of region E which contains a river and some tributaries. This is an example of which drainage pattern?
   A. annular
   B. dendritic
   C. parallel
   D. rectangular
   E. radial
   F. trellis

13. (2 points) Region D has a higher drainage density than region E (the two maps are not to the same scale). Provide two reasons for why this would be the case.

Solution: on D is steeper, region D has less vegetation, Region D has lower permeability

14. (1 point) True or false: Region D would have "flashier" floods

15. (1 point) Which of the following features forms at a nickpoint?

    A. Gilbert's Type Delta

    B. Stair-step Canyon

    C. Oxbow Lake

    D. Waterfall

16. (2 points) Select all statements which are NOT TRUE concerning base level

    A. Ultimate base level is sea level

    B. A lowering of base level results in faster retreat of a nickpoint upstream

    C. A raising of the base level results in migration of a nickpoint downstream

    D. A lowering of base level results in more downcutting

    E. A raising of base level results in more deposition

    F. A nickpoint is NOT a local base level.

3 Stream Geology, Round 2

1. (1 point) Which of the following sediments has the smallest diameter?

   A. clay

   B. gravel

   C. sand

   D. silt

2. (2 points) Order the following river zones from farthest to closest to the river mouth: Zone of erosion, Zone of transport, Zone of deposition

   Solution: Zone of erosion, Zone of transport, Zone of deposition

3. (2 points) A river has a velocity of 70 cm/second. Identify whether each of the following sediments will be eroded, transported, or deposited: Silt, Gravel, Cobbles

   Solution: Silt - Eroded. Gravel - Any of the 3. Cobbles - Deposited

4. (2 points) Now say sea level falls and the base level of this river changes. Would one expect to find the sediment load of this river to be higher or lower after this sea level change and why?

   Solution: Higher sediment load since lower sea level -¿ downcutting more erosion so more sediment

5. (3 points) Label the features on Station 3 Image A.

   Solution: A - Natural Levees B - Oxbow Lake C - Backswamp D - Yazoo Tributary E - Meander scar F = Meander Cutoff

6. (2 points) Refer to Station 3 Image A. Explain the formation of landforms D and C in the diagram. Would you expect to find these features towards the head of a river?

   Solution: Backswamp and yazoo tributaries formed bcuz water was trapped behind natural levees and can't come back to the main river. Wouldn't form more headward cuz only really forms with natural levees which is for meandering rivers so more headward is not meandering so no backswamps or yazoo tributary.

7. (2 points) Refer to Station 3 Image A. This meandering river is shown to have clear cut banks and point bars. Explain the formation of these two features and where they are located relative to the river bends

   Solution: Cut banks found on the outside of the river curve. Water has to travel farther distance so it moves faster so more erosion. Point bars are found on side of the river bend and since less distance needed to be traveled moves slower and more deposition of sediment

8. (1 point) Refer to Station 3 Image A. [True] or False - This river will migrate laterally

9. (2 points) List the 3 types of sediment loads in rivers and order them from smallest to largest in average sediment size

Solution: Dissolved Load, Suspended Load, Bed Load

10. (1 point) What sediment load is commonly moved through saltation?

Solution: Bed Load

11. (1 point) What sediment load transports the most sediment in most rivers?

Solution: Suspended Load

12. (1 point) What sediment is generally only transported during large floods?

Solution: Bed Load

4 Stream Flow and Flooding

Use the diagram of river channels on the image slideshow (Station 4 right image)for the following 3 questions. You may assume channel A is a perfect semicircle with river diameter of 8 meters and channel B is an isosceles trapezoid with a river base of 20 meters, a depth of 3 meters, and a surface length of 12 meters. Assume a Manning's roughness coefficient of 0.05 for both river channels and a river gradient of 0.02 meters per meter.

1. (6 points) Calculate the flow velocity for both rivers and determine which river has a faster flow velocity (Show your work :D)

   Solution: Channel A: Cross Sectional Area = $(4^2)/2\pi = 8\pi$ Wetted perimeter = $8/2pi = 4pi$ Hydraulic Radius = $2m$ Manning Equation: $(2 \cdot \sqrt{0.02})/0.05 = 4.489m/s$
   Channel B: Cross sectional area = $(12+20)/2x3 = 48m^2$ Wetted perimeter = $5+5+20 = 30$ (5 is from 3-4-5 triangle to get slant length) Hydraulic radius = $48/30$ Manning Equation: $0.8(1.6 \cdot \sqrt{0.02})/0.05 = 3.869m/s$
   River A faster

2. (2 points) Which river has the greater discharge? Show how you know thru some calculations

   Solution: River B. $48 \cdot 3.869 > 8\pi \cdot 4.489$

3. (3 points) Describe the difference between a Perennial, Intermittent, and Ephemeral stream.

   Solution: Perennial (flows all year round), usually groundwater fed. Intermittent (only flows some of the year). Ephemeral (only flows after rainfall)

4. (6 points) USGS has kept stats on the discharge and stage level of the Skokie River since 1986. This summer, you've decided to practice swimming in the Skokie River. You plan to swim every day of the summer (assume 3 full months / one quarter of a year. Assume this is 2022), but you don't want to swim on days when the stage is above or at seven feet. Since 1986, the Skokie River has matched or exceeded that stage 20 times. Assuming that the floods have been completely statistically average, what is the probability, rounded to the neared hundredth of a percent, you will be able to swim every day of the summer? Show your work.

   Solution: 87.03%
   Return period is $36/20 = 1.8$ yrs. Summer is 0.25 yrs.
   The probability of a certain flood not happening in a certain amount of years can be described as $100\frac{1}{e^{t/T}}$ where t is the time interval and T is the return period. You can plug t in as 0.25 and T in as 1.8 to solve for 87.03%.

5. (2 points) Now imagine that the watershed of the Skokie River undergoes major urbanization. Which of the following would be true about future floods of the Skokie river? Select all that apply.

   A. Flooding would be more frequent
   B. Floods would have a lower lag time
   C. Floods would have a higher discharge
   D. Floods would be slower

6. (5 points) Refer to the hydrograph on the image sheet (Station 4 left image).
   (a) (1 point) What type of graph is shown here?
   (b) (3 points) Label the graph for features A-E
   (c) (1 point) What is the approximate lag time of this storm?

Solution:
a) Hydrograph
b) A = Rain Discharge. B = Lag Time. C = peak discharge. D = Storm Flow. E = Base flow.
c) About 16 hours - anything reasonable

7. (2 points) Select all of the following that would influence the shape of a graph such as that in question 7

A. Storm Characteristics
B. Catchment size
C. drainage density
D. Soil Type

5 Groundwater

1. (1 point) True or [False]: Withdrawal of water from an aquifer always leads to lowering of the water table.

2. (1 point) [True] or False: A smaller sediment size typically correlates with a thicker capillary fringe

3. (1 point) True or [False]: A relatively low ratio of O-18 to O-16 in groundwater indicates that the groundwater was collected during a greenhouse earth period.

4. (1 point) Which of the following is an example of primary porosit?
   A. dissolution cracks in karst limestone
   B. the pore spaces in unconsolidated gravel
   C. fractures in a granitic sill
   D. joints in columnar basalt

5. (1 point) Which of the following sediments would have the lowest porosity
   A. glacial lodgement till (laid down directly under glaciers)
   B. glaciofluvial (laid down by glacial meltwater)
   C. eolian (laid down by wind)
   D. alluvium (laid down by running water)

6. (1 point) Artisan wells are wells from
   A. the vadose zone
   B. perched aquifers
   C. unconfined aquifers
   D. confined aquifers

7. (1 point) What are the two main components of a hydraulic head in a saturated aquifer?
   A. elevation head and tension head
   B. pressure head and gravity head
   C. pressure head and elevation head
   D. elevation head and constriction head

Refer to the subsequent information for the following questions. Storativity (S) is defined as the volume of water released from the storage per unit surface area per unit decline in hydraulic head. Specific Storage is defined as the volume of water released per decline in hydraulic head per unit volume of aquifer. Specific Yield is defined as the drainable porosity of an aquifer.

8. (1 point) Fill in the Blanks. Specific (storage/yield) is used to find the storativity of a confined aquifer and specific (storage/yield) is used to find the storativity of an unconfined aquifer.
   A. storage, storage
   B. yield, yield
   C. storage, yield
   D. yield, storage

9. (1 point) Assuming all of the following substrates have the same porosity, which of the following has the highest specific yield?
   A. unconsolidated sand

B. fractured metamorphic rock

C. clay

D. loam (a mix of sand, silt and clay)

10. (2 points) How would you determine the storativity of an aquifer from its specific storage?

Solution: multiply specific storage by thickness of aquifer

11. (2 points) Which of the following properties of an aquifer affect its Specific Storage? Select all that apply.

A. permeability

B. porosity

C. average grain size

D. pressure of water

E. compressibility of the aquifer

F. density of the water

12. (3 points) Hydrologists sometimes measure the concentration of CFCs in groundwater. What is the purpose of this and how can it be used for this purpose?

Solution: to age the groundwater (1pt), because CFC concentrations in the atmosphere has changed through time due to human addition to the atmosphere (1pt), concentration can be correlated to age (1pt)

13. (3 points) Two Piezometers are drilled into the ground 823 meters apart. The water rises to an elevation of 99m in the first piezometer and 93m in the second. How many days does it take for the water to travel the distance from Well 1 to Well 2 assuming the water travels in a straight line? Use a Saturated Hydraulic Conductivity of 0.65m/day. Provide your answer in scientific notation with 3 sig figs (ie 1.23E2). Show your work.

Solution:
(1)velocity (darcy's law): $0.65 \cdot \frac{99-93}{823} = 0.004739 m/day$
(1)distance (pythagorean theorem): $\sqrt{6^2 + 828^2} = 823.0218 m$
(1)time: $823.0218/0.004739 = 1.74 \cdot 10^5$days

14. (6 points) Refer to Station 5 Image A . This is a diagram of saltwater intrusion starting to take place. Answer the following questions:

(a) (2 points) What is the name of features A and why does it form?

(b) (3 points) What is the name of feature B and why does it form?

(c) (1 point) Why does the saltwater form a wedge under the freshwater, but the freshwater does not form a wedge under the saltwater?

Solution:
a) Cone of depression (1), excessive withdrawal(1)
b) Cone of ascension (1), less pressure from above because of cone of depression (1), so it rises (1)
c) Saltwater is denser(1)

6 Karst

1. (1 point) What term describes cave growth as a whole?

   Solution: Speleogenesis

2. (1 point) Which of the following is not a requirement for karst topography?
   A. Limestone terrain
   B. Excessive Sunlight
   C. Organic Soil
   D. High precipitation amounts

3. (1 point) Which of the following is false regarding phreatic passages?
   A. phreatic passages are wider than tall
   B. typically found along bedding planes
   C. have rounded/flattened ovoid profiles
   D. Often have a canyon-like notch at the base of the passage

4. (1 point) What structures are formed when a stalactite and a stalagmite meet and come together?

   Solution: column

5. (1 point) What term describes an inverted funnel-shaped sinkhole?
   A. Cenote
   B. Abime
   C. Doline
   D. Foibe

6. (1 point) What term describes the location where surface flow enters an underground system?
   A. Cenote
   B. Ponor
   C. Scowle
   D. Karst Window

7. (2 points) Explain how a karst region can have many different "layers" of caves on top of each other and how this leads to the development of sinkholes

   Solution: Karst regions follow with a water table. If water table drops or smth then another layer of caves can occur below and then very weak structure so can fall and sinkhole

8. (3 points) Write out the steps for the dissolution of limestone that occurs in Karst regions:

Solution:
$H_2O + CO_2 \rightarrow H_2CO_3$
$H_2CO_3 + CaCO_3 \rightarrow HCO_3^- + Ca^{2+} + CO_3^{2-}$

9. (2 points) Select all of the following which can be found in karst regions:

A. Sinkholes
B. Collapsed Breccia
C. Flowstone
D. Helictites
E. Aretes

7 Lake Geology

1. (1 point) Which of the following types of lakes is most likely to become salty?
   A. Volcanic Lake
   B. Playa Lake
   C. Fluvial Lake
   D. Solution lake

2. (1 point) Which of the following lake types is generally the deepest?
   A. sag ponds
   B. caldera lakes
   C. kettle lakes
   D. graben lakes

3. (1 point) Which of the following lakes is most susceptible to a GLOF?
   A. proglacial lake
   B. epiglacial lake
   C. supraglacial lake
   D. subglacial lake

4. (2 points) A glacial lake that forms in a circular depression (cirque) from the scouring action of glaciers is known as a: _____

   Solution: tarn

5. (1 point) Refer to Station 7 Image A which shows a topographic image of Lake Winnipesaukee, a non anthropogenic lake. In which of the following ways did it most likely form?
   A. glacial scouring
   B. damming action of a moraine
   C. tectonic rifting
   D. caldera collapse
   E. inundation with water due to overgrowth of sphagnum moss

6. (1 point) Which of the following statements regarding Lake Winnipesaukee is false
   A. Lake Winnipesaukee acts as a local base level for some adjacent rivers
   B. Lake Winnipesaukee receives minimal fertilizer runoff because the adjacent land is too steep for farming
   C. Lake Winnipesaukee is very deep
   D. Lake Winnipesaukee is exorheic

8 Lake Chemistry

1. (2 points) Explain the difference between Endorheic and Exorheic lakes

Solution: Exorheic lakes into a river, or other body of water that ultimately drains into the ocean. Endorheic lakes has water that does not drain into the ocean, but are reduced by evaporation, and/or drain into the ground

2. (1 point) Which of the following is false regarding oligotrophic lakes?
   A. Generally younger lakes
   B. Low in nutrients
   C. high in organic matter
   D. Fairly deep

3. (1 point) Which of the following would be most useful in determining the mixing pattern of a river?
   A. monthly temperature profile
   B. monthly nutrient profile
   C. monthly oxygen profile
   D. monthly pH profile

4. (1 point) Which of the following is true regarding lakes?
   A. Lakes can generally transition from oligotrophic to Eutrophic but NOT Eutrophic to Oligotrophic
   B. Lakes can generally transition from Eutrophic to Oligotrophic but NOT Oligotrophic to Eutrophic
   C. Lakes cannot transition between these types
   D. Lakes can transition between Oligotrophic and Eutrophic interchangeably

5. (5 points) Refer to the diagram on station 8 of the image slideshow (LAKE CHEMISTRY) for this question.
   (a) (1 point) Label Layers I, II, and III
   (b) (2 points) This lake is dimictic. What season is it in, and why?
   (c) (1 point) In which layer would you expect the lowest oxygen concentration and why?

Solution:
a) I = epilimnion II = metalimnion III = hypolimnion.
b) Most likely summer cuz thermal stratification
c) Epilimnion or I layer would most likely have lowest oxygen concentration because respiration would use oxygen plus warmer = holds less dissolved gases

6. (3 points) Explain what a meromictic lake is and some possible ecological threats such a lake poses to the organisms living inside of it and around it.

Solution: a lake which has layers of water that do not intermix. No mixing means monimolimnion or the bottom layer of the lake circulates very little, and are generally hypoxic and saltier than the rest of the lake. ALSO Limnic eruptions when mixing does occur and the stored gas like carbon dioxide in the bottom layers escape out into atmosphere and can kill a lot of nearby organisms and people

7. (2 points) Explain what a dimictic lake is and in what seasons do the layers mix in such a lake?

Solution: A lake which is stratified during winter and summer but can mix during fall and Spring.

8. (2 points) Explain the stratification in Amictic lakes. Stratification in these lakes is most similar to what season in dimictic lakes?

Solution: Amictic lakes exhibit inverse cold water stratification whereby water temperature increases with depth below the ice surface. Most similar to winter in dimictic lakes

9. (2 points) What does TDS and DO stand for in regards to lake chemistry?

Solution: TDS = Total dissolved SOlids. DO = Dissolved Oxygen

9 Wetlands and Critical Zone Hydrology

1. (2 points) What is the definition of a wetland

   Solution: an area of land that is saturated for at least part of the year, has hydric soils and a special vegetation time

2. (3 points) Environmentalists are concerned about the shrinking and disappearance of wetlands around the world. This is because of their profound benefits on Earth's environment and ecology. Give 3 examples of the benefits of wetlands.

   Solution: filter out pollutants, recharge, stormwater/flood control, habitat for species, anything reasonable

3. (1 point) Which of the following types of wetlands is wet for only part of the year and includes dormant trees (oaks, willows, bald cypress) and is dominated by woody plants?
   - **A. Swamp**
   - B. Bog
   - C. Marsh
   - D. Pocosin
   - E. Fen

4. (1 point) Which of the following types of wetlands is wet the entire year and only contains grasses (cattails, bulrushes, lilies)
   - A. Swamp
   - B. Bog
   - **C. Marsh**
   - D. Pocosin
   - E. Fen

5. (1 point) Which of the following types of wetlands has evergreen shrubs and trees. No standing water, but a shallow water table with soil saturated for most of the year
   - A. Swamp
   - B. Bog
   - C. Marsh
   - **D. Pocosin**
   - E. Fen

6. (7 points) Refer to Station 9 Image A for the following questions:
   (a) (1 point) What type of wetland is depicted?
   (b) (1 point) What are these types of wetland fed by?
   (c) (2 points) These wetlands can form in 2 major ways, what are these 2 processes?
   (d) (2 points) Does organic material dissolve quickly in this wetland? Why or why not? This allows what to form in this wetland?
   (e) (1 point) Are these wetlands acidic? Provide one reason for why or why not?

Solution:
a) Bog
b)Only precipitation
c) moss grows over a lake or pond and slowly fills it (terrestrialization) b. sphagnum moss blankets dry land and prevents water from leaving the surface (paludification)
d) water doesn't flow in and out of them easily. Low levels of oxygen and cold temperatures make it more difficult for fungi and bacteria to decompose dead plants quickly. This helps peat form.
e)Yes, very acidic. possible reasons: anearobic decomposition releases acidic H2S, only precip fed and rain is slightly acidic, no groundwater buffers such as carbonate ions, sphagnum moss makes things acidic through ion exchange.

7. (1 point) True or [False], bogs are more common in warm tropical climates than cooler climates

8. (1 point) True or [False], all wetlands are acidic.

9. (2 points) A layer of soil that freezes for more than 15 days per year but is not frozen year round is called _____ (3 words).

Solution: seasonally frozen ground

10. (1 point) Which of the following is false regarding open and closed pingos?
    A. Open systems receive liquid water continuously from groundwater aquifer
    B. In a closed system, confined mass of saturated soil freezes and expands, pushes on overlying material
    C. Closed system has hydrostatic pressure while open system has hydraulic pressure
    D. A closed system has an underground ice lens, an open system has a surface exposed ice lens